**Memo: Customers Like You**

**To:** Sarah
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**Subject:** PCA vs. SVD for Product Recommendations

## Introduction

In this project, I used the Online Retail II dataset to understand customer purchasing patterns and see how we can recommend products based on previous purchases. I cleaned the data, divided it into training and test periods based on dates, and created a customer-product matrix. I then used PCA and SVD to find patterns in the data and compared their performance.

## What I Found

I tested both PCA and SVD with 10, 20, 50, 100, and 200 components. The best-seller baseline achieved a Hit Rate@10 of 17.41%. With 200 components, PCA achieved 40.04% and SVD achieved 39.94%. This means both methods performed better than the baseline in my evaluation.

The results from PCA and SVD were very similar. Both explained around 57.4% of the variance at 200 components, and their reconstruction errors were almost the same. However, SVD was faster in my experiment, taking about 1.30 seconds compared with 1.97 seconds for PCA.

I also found that the customer-product matrix was very sparse, with about 97.95% of its entries being zero. Storing it as a sparse matrix required around 4.76 MB, compared with 154.36 MB for a dense matrix. This shows why sparse data handling is useful for this type of problem.

I also explored five customer groups based on their purchasing patterns. These groups included customers interested in games and gifts, bags and storage, decorative homeware, baking products, and customers who bought a wider variety of products less frequently.

## Conclusion

Based on my results, I would choose SVD as a practical option because it was faster in my experiment and works directly with sparse data. PCA also performed well, and the difference in recommendation performance between the two methods was very small.

There are some limitations to my work. I used the same held-out data to compare different component counts, so the results need further testing. Also, the customer groups were not clearly separated, as shown by the low silhouette score of approximately 0.0173. I would therefore treat them as initial observations rather than definite customer types.

Overall, this project helped me understand how PCA and SVD can be used to study customer purchase patterns and how recommendation quality, processing time, and customer segmentation need to be considered together.